
Hicham Lamrani

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Personal Summary

Industrial & Systems Engineering and Data Science graduate student seeking an internship in Data Analytics and Machine Learning – available Jan 2026.

Education

Polytech Clermont - France | Master of Science

Mathematical Engineering And Data Science, Expected in 12/2026

University of Oklahoma - Norman, OK | Master of Science

Industrial And Systems Engineering, Expected in 12/2025

Polytech Clermont - France | Bachelor of Science

Mathematical Engineering And Data Science, 06/2023

Experience

COST ENGINEERING PROJECT – WAREHOUSE AUTOMATION | 06/2025 - 08/2025

University of Oklahoma - Norman, Oklahoma

- Conducted a 10-year financial evaluation of warehouse automation (robotics, WMS, conveyors) vs. manual operations
- Applied core tools: **NPV, IRR, ROI, payback period, sensitivity analysis, and Monte Carlo simulation**
- Developed a **Job Costing Ledger** and performed a **Joint Confidence Level** risk assessment (10,000 iterations)
- Delivered a professional project poster and financial model using **Excel and Python**
- **Key Results:** NPV \$1.38M | ROI 290% | Payback: 3 years

EARLY ICU MORTALITY PREDICTION (MIMIC-III DATASET) | 01/2025 - 05/2025

University of Oklahoma - Norman, Oklahoma

- Built and evaluated ML models (**XGBoost, LightGBM, logistic regression**) to predict ICU mortality using early clinical data from the **MIMICIII** database
- Designed an interpretable pipeline with **robust preprocessing, hyperparameter tuning, and SHAP explainability**
- Achieved **AUC = 0.90**, outperforming traditional scoring systems (e.g., SAPS II and prior benchmarks)
- Contributed to the development of **reproducible, clinically actionable** decision-support tools for critical care settings

INVESTIGATING THE IMPACT OF VISUAL CONSPICUITY ON | 01/2025 - 05/2025

University of Oklahoma - Norman, Oklahoma

- Designed and conducted a **controlled behavioral experiment** to study the effects of **stimulus visibility and prior exposure** on inattentional blindness
- Developed custom experimental videos (inspired by the **Simons & Chabris "gorilla" study**) manipulating **color**

- contrast** under high cognitive load
- Performed statistical analysis using **logistic regression** and **Chi-square tests**; found significant effects on detection likelihood (**AUC = 0.76**)
- Contributed to **human factors research** with implications for visual design in **safety-critical environments** (e.g., driving, healthcare)

DATA ANALYSIS INTERN | 03/2024 - 08/2024

University of Malmö - Malmö, Sweden

- Analyzed multi-sensor data (wearables, smartphones, Kinect) from physical performance tests to quantify symptoms in cardiac and neurological disorders.
- Developed an interactive Dash app for data visualization and movement analysis.
- Built a 2D skeleton animation tool and contributed to a published research paper.

DATA SCIENCE PROJECT | 09/2023 - 03/2024

Polytech Clermont - France

- Successful Design and Early Delivery of an Automated Dashboard, Analyzing Relationships and Performance of Companies Engaged in Agro-ecological Transition at Kèpos. Accompanied by a Manual and In-depth Scientific Study on Collaborations and Their Impacts.

Skills

- | | |
|--|---|
| <ul style="list-style-type: none">PythonR programmingSQLC and C++JavaAMPLPower BIMachine Learning | <ul style="list-style-type: none">Deep LearningData VisualizationStatistical analysisProblem solvingTeam collaborationPredictive modelingData cleaningProject management |
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Languages

English:



Spanish:



French:



Arabic:

